 # Task1

```
import tkinter as tk

def convert():
    c = float(entry.get())
    f = (c * 9/5) + 32
    result.config(text=f"Fahrenheit: {f}")

root = tk.Tk()
root.title("Temperature Converter")

entry = tk.Entry(root)
entry.pack()

btn = tk.Button(root,text="Convert",command=convert)
btn.pack()

result = tk.Label(root,text="")
result.pack()

root.mainloop()
```

 # Task2

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt

nums = np.random.randint(1,100,20)
df = pd.DataFrame(nums,columns=["Numbers"])

print("Mean:",np.mean(nums))
print("Median:",np.median(nums))
print("Std Dev:",np.std(nums))

plt.hist(nums)
```

```
plt.title("Random Number Histogram")
plt.show()

# Task3
class Book:
    def __init__(self,title,author,price):
        self.title = title
        self.author = author
        self.price = price

    def display(self):
        print(self.title,self.author,self.price)

b1 = Book("Python","Guido",500)
b1.display()

# Task4
class Product:
    def __init__(self,name,price):
        self.name=name
        self.price=price

class Cart:
    def __init__(self):
        self.items=[]

    def add(self,product):
        self.items.append(product)

    def total(self):
        t=0
        for i in self.items:
            t+=i.price
        print("Total:",t)
```

```
p1=Product("Laptop",50000)
p2=Product("Mouse",500)

cart=Cart()
cart.add(p1)
cart.add(p2)
cart.total()
```

 # Task5

```
class ATM:
    def __init__(self,balance):
        self.balance=balance

    def check(self):
        print("Balance:",self.balance)

    def deposit(self,amt):
        self.balance+=amt

    def withdraw(self,amt):
        self.balance-=amt

atm=ATM(1000)
atm.deposit(500)
atm.withdraw(200)
atm.check()
```

 # Task6

```
class Student:
    def __init__(self,name):
        self.name=name

class Teacher:
    def __init__(self,name):
        self.name=name
```

```
class Course:
    def __init__(self,name,teacher):
        self.name=name
        self.teacher=teacher

    def display(self):
        print(self.name,self.teacher.name)

t=Teacher("Mr Rao")
c=Course("Python",t)
c.display()
```

 # Task7

```
for num in range(1,101):
    prime=True
    if num>1:
        for i in range(2,num):
            if num%i==0:
                prime=False
                break
        if prime:
            print(num)
```

 # Task8

```
s="ABCDEFGHIJKL"
print(s[2:9:2])
print(s[10:2:-1])
print(s[10::-1])
print(s[10:1:-2])
print(s[0:12:4])

t="Python String Slicing Example"
print(t[::-1])
print(t[7:])
print(t[2::3])
```

```
print(t[0::2])
print(t[-7:])
print(t[::4])
```

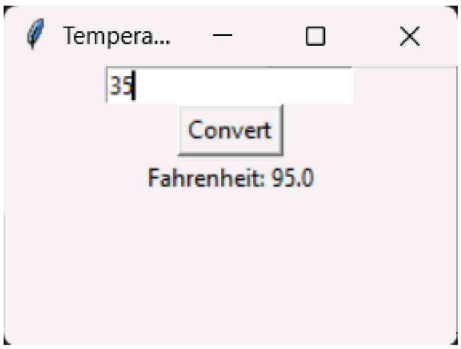
 # Task9

```
lst=["Cat","Dog","Lion","Tiger","Rabbit","Monkey"]

print([lst[2]])
print([lst[5],lst[4]])
print([lst[3],lst[2],lst[1]])
print([lst[0],lst[3]])
print([lst[3],lst[0]])
print([lst[5],lst[2]])
print([lst[4],lst[2],lst[0]])
print(lst[::-1])

l=["apple","banana","cherry"]
l.append("orange")
print(l)

l.insert(1,"mango")
print(l)

l.extend(["kiwi","grape"])
print(l)

n=[10,20,30,40,50]
n[2]=300
print(n)

n=[10,20,30,40,50]
n[1]=200
n[2]=3000
n[3]=400
print(n)
```

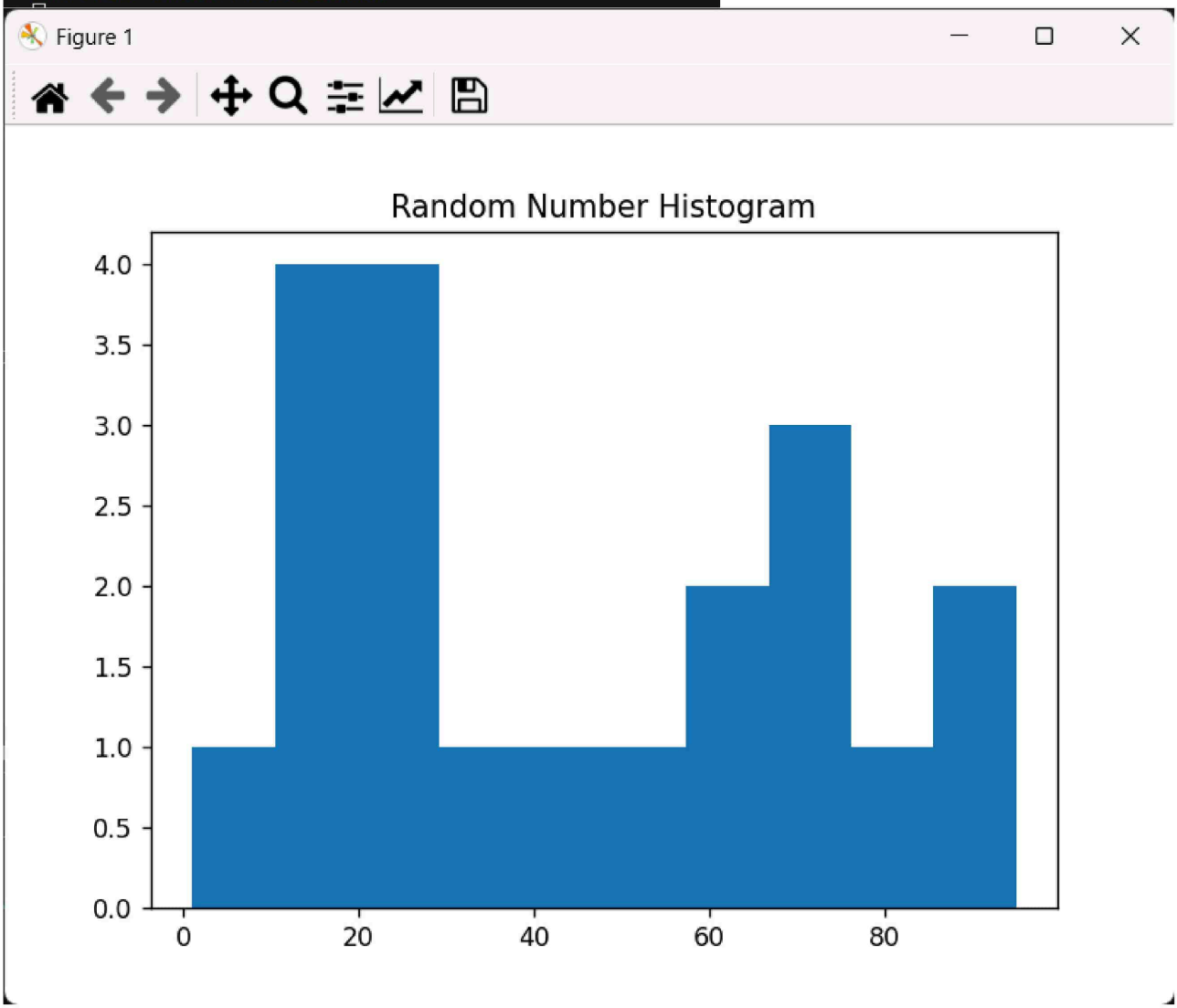
```
a=[1,2,3]
a.insert(1,100)
print(a)

a=[1,2,3]
a.insert(1,100)
a.append(999)
print(a)

b=[10,20,30,40,50]
b.append(60)
print(b)

b.insert(0,5)
print(b)

b.extend([70,80,90])
print(b)

c=[42,3.14,"Hello",True]
c[0]=2.718
print(c)

c.append(1000)
print(c)

c.insert(1,False)
print(c)

c[0]=5
print(c)

txt="Hello World, Welcome to Python!"
print(txt.upper())
print(txt.lower())
```

```
print(txt.split(" "))
print(txt.split("o"))
print(txt.replace("W","X"))
```

Outputs:




```
(.venv) PS D:\Internship\Day36> python task1-9.py
Mean: 43.75
Median: 35.0
Std Dev: 28.442705567508867
```



```
Python Guido 500
Total: 50500
Balance: 1300
Python Mr Rao
2
3
5
7
11
13
17
19
23
29
31
37
41
43
47
53
59
61
67
71
73
79
83
89
97
```

```
String : ABCDEFGHIJKL

CEGI
KJIHGFED
KJIHGFEDCB
KIGE
AEI

String : Python String Slicing Example

gnirtS nohtyP
Slicing Example
emEni iSoy
Potgigae
elpmaxE
gtoP

List : ['Cat', 'Dog', 'Lion', 'Tiger', 'Rabbit', 'Monkey']

Lion
Monkey Rabbit
['Tiger', 'Lion', 'Dog']
['Cat', 'Tiger']
['Tiger', 'Cat']
['Monkey', 'Lion']
['Rabbit', 'Lion', 'Cat']
['Monkey', 'Rabbit', 'Tiger', 'Lion', 'Dog', 'Cat']
```

```
List : ['apple', 'banana', 'cherry']

['apple', 'banana', 'cherry', 'orange']
['apple', 'banana', 'cherry', 'cherry', 'orange']
['apple', 'banana', 'cherry', 'cherry', 'orange', 'kiwi', 'grape']

List : [10, 20, 30, 40, 50]

[10, 20, 300, 40, 50]
[10, 200, 3000, 400, 50]

List : [1, 2, 3]

[1, 100, 2, 3]
[1, 100, 2, 999]

List : [10, 20, 30, 40, 50]

[10, 20, 30, 40, 50, 60]
[5, 10, 20, 30, 40, 50, 60]
[5, 10, 20, 30, 40, 50, 60, 70, 80, 90]

List : [42, 3.14, 'Hello', True]

[2.718, 3.14, 'Hello', True]
[2.718, 3.14, 'Hello', True, True]
[2.718, False, 3.14, 'Hello', True, True]
[5, 3.14, 'Hello', True, True]

String : Hello World, Welcome to Python!

HELLO WORLD, WELCOME TO PYTHON!
hello world, welcome to python!
['Hello', 'World,', 'Welcome', 'to', 'Python!']
['Hell', ' W', 'rld, Welc', 'me t', ' Pyth', 'n!']
Hello Xorld, Xelcome to Python!
```